

Packet Scheduling in Routers

1 Introduction

Packet scheduling determines the order in which queued packets are transmitted over an outgoing link. This is similar to how humans wait in line at service centers, bus stops, banks, or airports.

Packet scheduling = selecting which waiting packet gets the link next.

Routers use a variety of scheduling disciplines, including:

- First-In-First-Out (FIFO)
- Priority Queuing
- Round Robin

Each discipline has different fairness, delay, and service characteristics.

2 First-In-First-Out (FIFO)

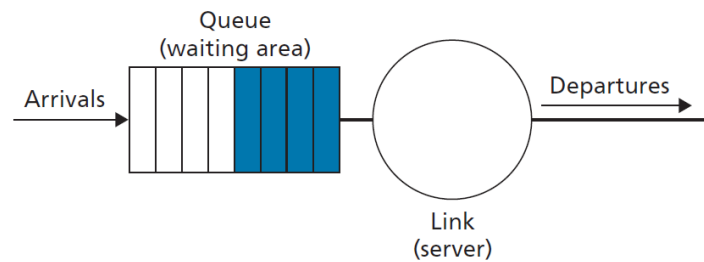
FIFO (or FCFS — First-Come First-Served) is the simplest and most widely used scheduling rule.

2.1 How FIFO Works

FIFO transmits packets in the exact order they arrive in the queue.

If the link is busy transmitting another packet, incoming packets wait in the queue. If the buffer overflows, packets are dropped based on the discard policy.

2.2 FIFO Queue Structure

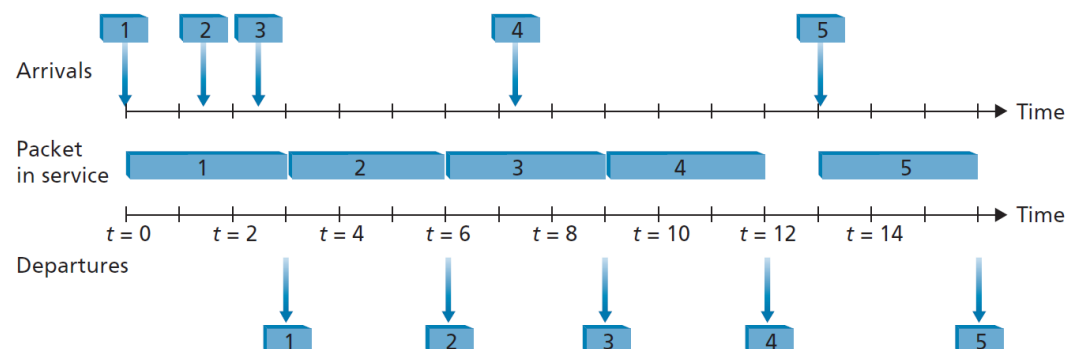


2.3 Example Operation

Assume each packet takes ****3 time units**** to transmit.

Packet arrival order: 1, 2, 3, 4, 5 Under FIFO, departure order will also be: 1, 2, 3, 4, 5.

Figure below represents arbitrary timeline behavior:



If no packets arrive after 4, the link remains idle until packet 5 arrives.

3 Priority Queuing

Priority queuing classifies arriving packets into separate priority classes.

3.1 Basic Idea

Higher-priority queues are always served before lower-priority queues.

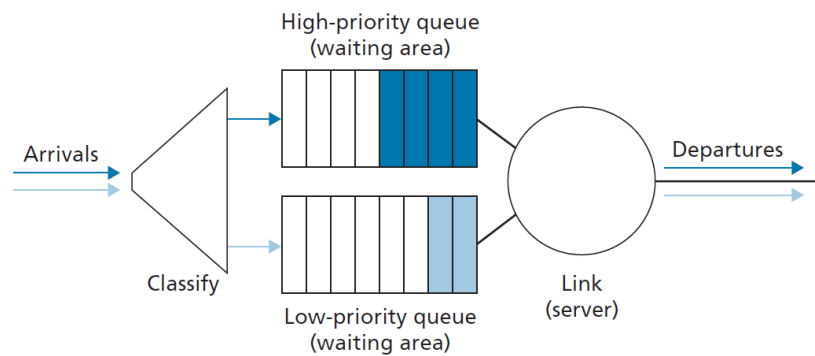
Routers often assign:

- Highest priority $\rightarrow$ Network control traffic or VoIP

- Lower priority → Regular user traffic (web, email, file transfers)

Packets in the same priority class are usually served FIFO.

3.2 Priority Queue Architecture



There is:

- A classifier that assigns each packet a priority
- A separate queue for each priority level
- A scheduler that transmits the highest-priority nonempty queue

3.3 Non-Preemptive Priority Scheduling

Most router priority scheduling is **non-preemptive**:

Once the router starts transmitting a packet, it will finish it — even if a higher-priority packet arrives.

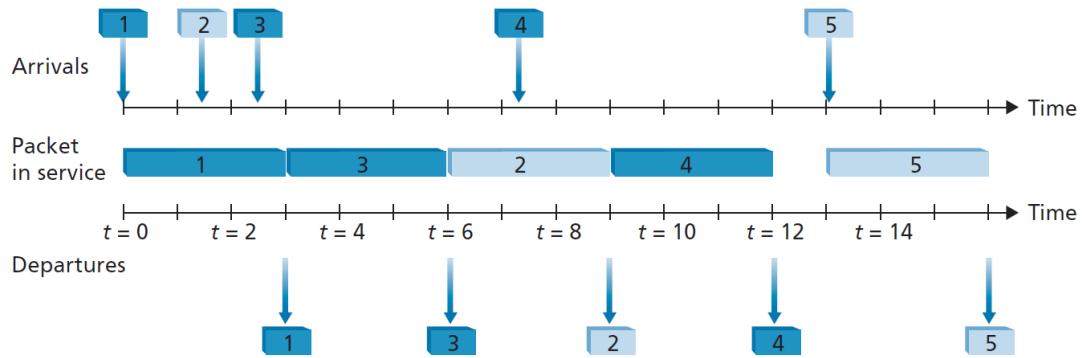
3.4 Example Timeline

Packets: 1, 3, 4 = high priority 2, 5 = low priority

Transmission order: 1 → 3 → 2 → 4 → 5

(3 jumps ahead of 2 because it is higher priority.)

Diagram representation:



4 Round Robin Scheduling and Weighted Fair Queuing (WFQ)

4.1 Round Robin Queuing

Round Robin is a fair scheduling technique in which packets are grouped into classes (similar to priority queuing), but ****no class has strict priority**** over another.

Round Robin: The scheduler cycles through classes in order, giving each class one opportunity to send one packet per cycle.

4.2 Work-Conserving Behavior

A round-robin discipline is called **work-conserving** if:

- The link is **never idle** when packets are waiting.
- If a class has no packets, the scheduler **immediately checks the next class**.

This ensures efficient link utilization.

4.3 Example Operation (Two Classes)

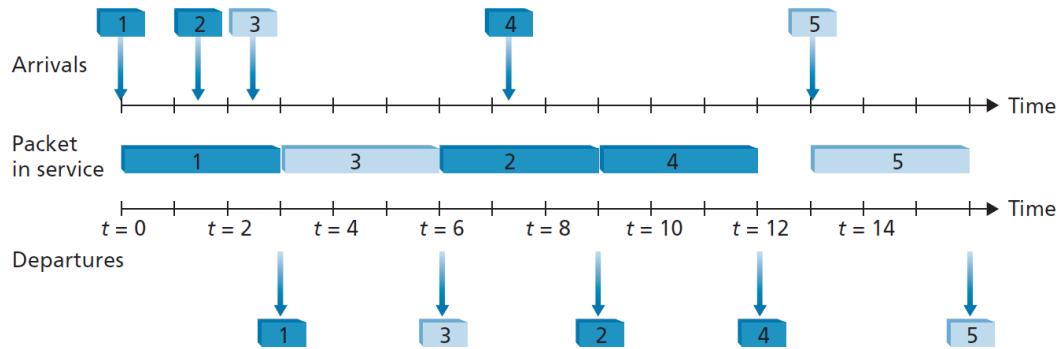
Packets belong to two classes:

Class 1: 1, 2, 4 Class 2: 3, 5

Transmission order: 1 (C1) → 3 (C2) → 2 (C1) → 4 (C1) → 5 (C2)

Explanation:

- Packet 1 arrives, link is idle $\rightarrow$ transmitted immediately.
- Packets 2 (C1) and 3 (C2) queue during packet 1's transmission.
- Scheduler alternates: Class 1 $\rightarrow$ Class 2 $\rightarrow$ Class 1 $\rightarrow$...
- Packet 4 arrives while Class 1 is being served next.



4.4 Weighted Fair Queuing (WFQ)

Weighted Fair Queuing is a more advanced and generalized form of round robin. It is widely implemented in modern routers.

WFQ allows classes to receive different proportions of bandwidth based on assigned weights.

4.5 Basic Mechanism

- Packets are classified into multiple classes (flows).
- Each class i is assigned a weight w_i .
- Scheduler cycles through classes, but the **amount of service** per class depends on its weight.

Unlike basic Round Robin:

WFQ guarantees bandwidth share proportional to the assigned weight, even under congestion.

4.6 Service Guarantee

During any interval of time when class i has queued packets:

Class i gets at least $\frac{w_i}{\sum w_j}$ of the link bandwidth.

Where the denominator is the sum of weights of all classes that currently have packets waiting.

If all classes have packets, and the link rate is R :

$$\text{Throughput of class } i = R \cdot \frac{w_i}{\sum w_j}$$

Thus:

- Higher-weight classes get more bandwidth.
- Lower-weight classes still get fair minimum service.
- No class is completely starved.

4.7 Packetization Issue

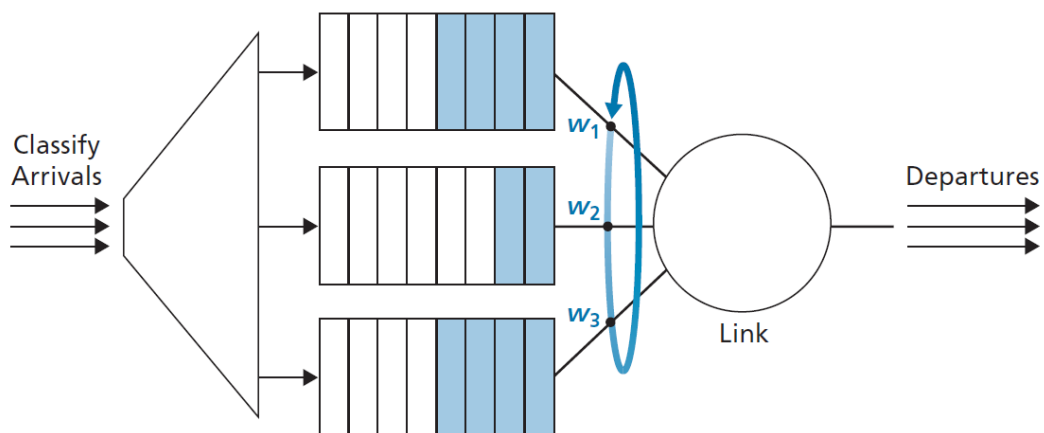
Real routers transmit whole packets and do not interrupt a packet midway.

Therefore:

WFQ emulates “bit-by-bit fairness” but implements it using whole packets.

(Referenced by Demers (1990) and Parekh (1993).)

4.8 WFQ Architecture



This model resembles round robin but uses weighted service and better fairness guarantees.

Comparison Summary

- **Round Robin:** Equal turn-taking among classes. Simple fairness.
- **WFQ:** Provides **weighted bandwidth sharing**, guarantees minimum throughput, prevents starvation.
- **Both:** Work-conserving $\rightarrow$ never leave the link idle when packets exist.

Summary

- **FIFO:** Simple, fair arrival order, but can suffer head-of-line blocking.
- **Priority Queuing:** Ensures low delay for high-priority traffic but may starve low priority.
- **Round Robin:** Fair sharing among multiple queues; prevents starvation.

Routers choose scheduling disciplines based on desired QoS and fairness properties.